

MeshPay: Resilient Offline Payment with Wireless Mesh Network

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December 19, 2025

Methodology

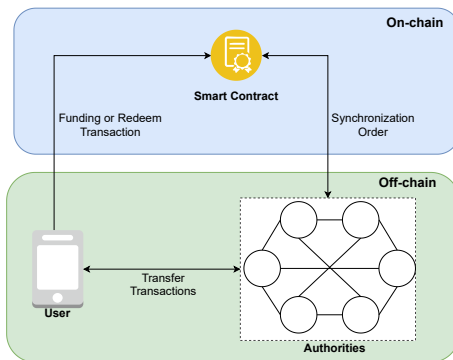


Figure: Transfer of funds between Blockchain and Protocol

Our Solution Architecture

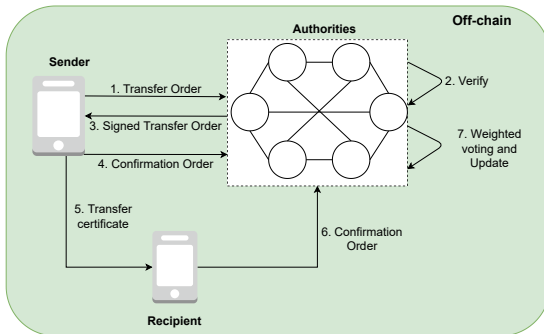


Figure: Transfer of funds between Users

Evaluation: Experimental Setup

We evaluate MeshPay using an enhanced Mininet-WiFi emulator configured to model a realistic mobile wireless mesh network.

Parameter	Value
Area	1000 × 1000 m
Number of users	50
Number of authorities	5–20
Mesh SSID	meshNet
Order interval	10 s
Propagation loss	logDistance
Mobility model	Random Direction
Transmission range	20 m
Node speed	0–20 m/s

Evaluation: Metrics & Comparison

Metrics

- **Latency:** Time from issuing a transfer order to collecting $\geq 2/3$ weighted signatures.
- **Throughput:** Number of finalized transactions per second.

Comparison Models

- **MeshPay-Weighted:** Proposed adaptive weighted voting.
- **MeshPay-Normal:** PBFT-style protocol requiring $2f + 1$ signatures.

We evaluate performance under:

- 1 Ideal network (no faulty authorities)
- 2 Byzantine faults
- 3 Varying user density

Results: Ideal Network Conditions

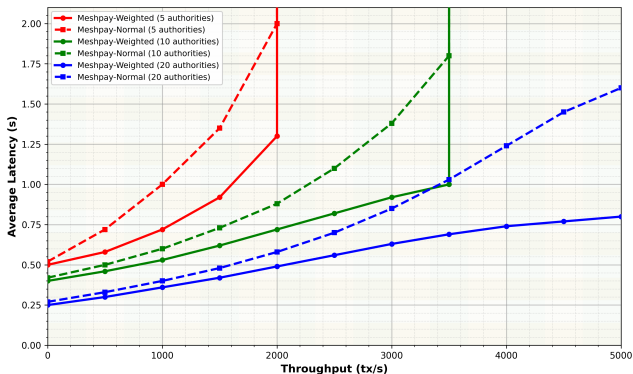


Figure: Throughput-latency under ideal conditions

Results: Performance Under Faults

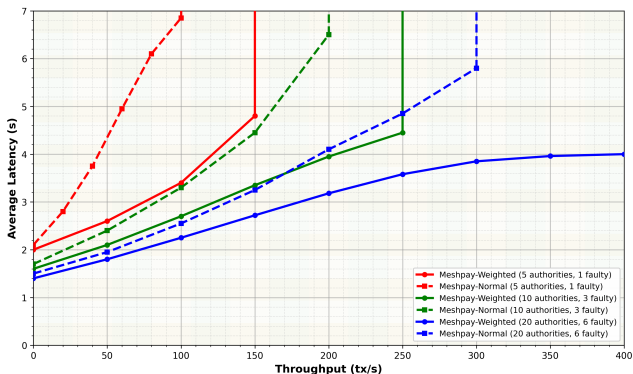


Figure: Throughput-latency with Byzantine nodes

Results: Impact of User Density

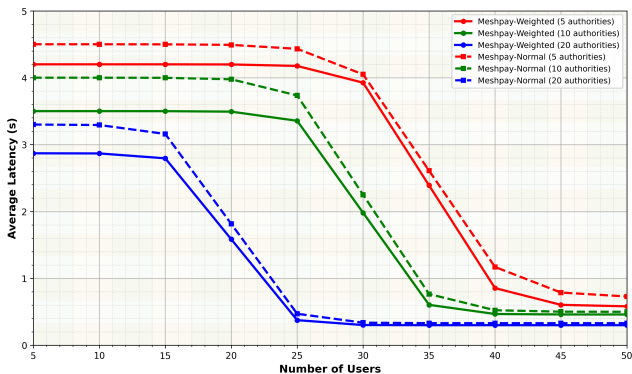


Figure: Latency with number of user nodes

Limitations

- **Dependence on authority reachability:** Our potocol requires quorum (2/3) weighted signatures — high latency if authorities are partitioned or multi-hop delays are high.
- **Sequential Bottlenecks:** Traditional blockchains depend on a linear sequence. In rural multi-hop meshes, the confirmation latency is bottle-necked by the pessimistic delay upper bound Δ , causing the system to stall during "view-changes" if a leader moves out of range.
- **Redundant Broadcasts:** Broadcasts to all authorities duplicate data across constrained wireless links, leading to increased packet collisions and rapid energy depletion of constraint-powered nodes.

Limitations

- **Missing Control Plane:** In decentralized wireless mesh networks, the absence of a coordinated control plane results in a lack of global network knowledge, making routing and message delivery extremely complex due to intermittent connectivity and dynamic topology.

Proposed Architecture

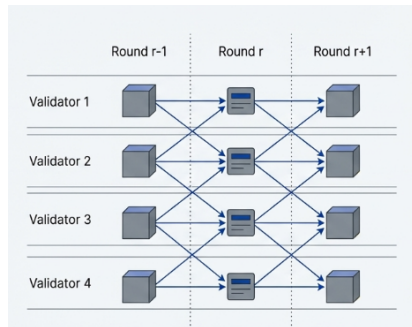
Decoupled Data & Control Plane for Rural Resilience

- **DAG-based Async Consensus:** Implements a round-based DAG where latency is a function of actual network delay, not Δ . This enables parallel transaction processing and "Zero-message" ordering.
- **Adaptive $RS(k, m)$ Erasure Coding:** Integrates an availability layer that fragments data. A quorum certificate proves availability of fragments (not full payloads), reducing per-link load and enabling reconstruction from k reachable chunks.
- **Decentralized SDN Orchestration:** Nodes elect temporary mobile controllers (drones/vehicles) to push FRL policies.

What is DAG?

Directed Acyclic Graph (DAG)

- A DAG is a graph of nodes connected by directed edges with **no cycles**.
- In blockchain and consensus systems, each node represents a **block, transaction batch, or vote**.
- DAGs enable **parallel proposal, high throughput, and low latency** by allowing multiple blocks to be appended simultaneously.
- Modern BFT systems (Narwhal, Tusk, Mahi-Mahi, Avalanche, Sui) leverage DAGs for **asynchronous ordering**.

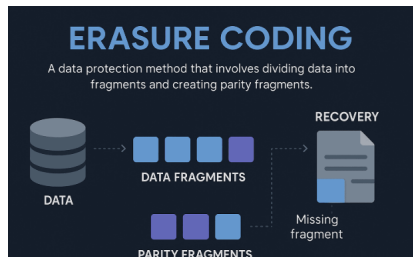


DAG-based Blockchain

What is Erasure Coding (EC)?

Redundancy without Replication

- **Definition:** A method of data protection that fragments a block into k data shards and m parity shards ($RS(k, m)$).
- **Property:** The original data can be reconstructed from *any* k fragments, even if m shards are lost or corrupted.
- **How we use it:** Instead of every node storing a full copy of the ledger, nodes store only one small fragment (size $1/k$). This results in storage costs from $O(N)$ to $O(1)$.

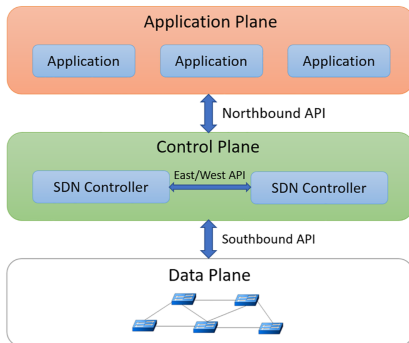


Erasure Coding

Decentralized SDN

Programmable Multi-Transport Overlays

- **SDN:** a network architecture that separates network control (the "brain") from forwarding (the "muscle"), allowing IT to manage the network centrally using software and APIs instead of manual hardware configurations.
- **How we use it:** We elect temporary "Mobile Controllers" in the mesh (e.g., a mobile phone, a vehicle or drone) to collect topology data and push routing policies learnt from FRL.



Decentralized SDN.

Adaptive Erasure-Coded DAG

Proposed innovation: *Adaptive Erasure-Coded (AEC)* — dynamically choose an $RS(k, m)$ code where

$$k = N' - F,$$

with N' = estimated number of currently available nodes and F = target fault tolerance. Each batch is split into k data + m parity fragments; any k fragments suffice to reconstruct [zhang2023minimizing].

Theoretical foundation:

- Model N' as a stochastic variable (nodes honest/faulty); predict via two-layer Bayesian smoothing on heartbeat / encounter traces.
- Choose k to minimize the expected storage per node subject to $\Pr(\text{recoverable}) \geq 1 - \delta$.
- Analyze recovery probability: for random fragment placement and independent availability p ,

$$\Pr(\text{recoverable}) = \sum_{i=k}^{k+m} \binom{k+m}{i} p^i (1-p)^{k+m-i}.$$

Adaptive Erasure-Coded DAG (cont.)

Evaluation targets (metrics):

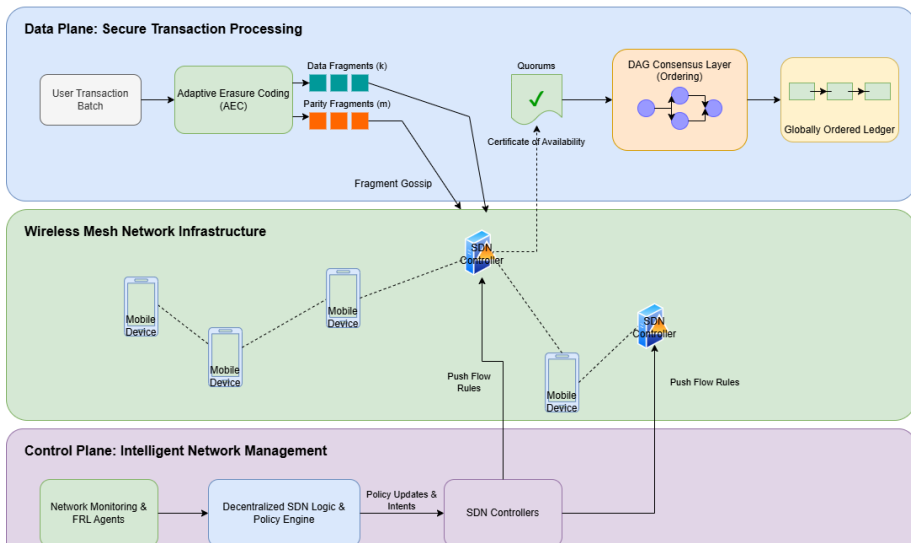
- *Storage redundancy*: $\frac{\text{bits stored across mesh}}{\text{original bits}}$.
- *Commit latency*: time from submission to certified DAG vertex (availability certificate).
- *Energy per commit*

DAG-Synchronized SDN Orchestration

- **Architecture:** Disaggregated SDN controllers synchronized via the DAG "Knowledge Plane." This ensures a consistent global view for flow-rule enforcement even during partitions.
- **Programmable Logic:** Injects knowledge from FRL and network monitoring into the control logic to dynamically reconfigure cluster definitions and reroute flows.
- **Failover Logic:** Employs a pre-partitioning strategy for master-slave handover between mobile controller, reducing failover latency to $< 200\text{ms}$.

Architecture Overview

Architecture Overview: DAG-based Mesh Network with Decentralized SDN & FRL



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